1. Clinical Question & Core Hypothesis

Clinical Question

Triple-negative breast cancers (TNBC) are classically HER2-negative and thus excluded from trastuzumab therapy, yet clinical and molecular data suggest subsets with low/heterogeneous ERBB2 expression might still engage HER2-directed mechanisms — why do some ERBB2-expressing TN tumors fail to benefit from trastuzumab while others might respond? 2) Tier 1 path used: the provided path links 'triple-negative phenotype → outcome → trastuzumab → HER2/ERBB2 → breast tissue → breast invasive carcinoma'; I built on ERBB2 as the molecular anchor because trastuzumab efficacy is mechanistically tied to ERBB2 expression. 3) Tier 2/supporting logic: immune effector engagement (ADCC) is a known mechanism of trastuzumab activity, so spatial immune context (captured by Tumor-Infiltrating Lymphocyte density on H&E) plausibly modulates whether ERBB2-low TN tumors derive benefit. 4) Endpoint choice: I selected therapy_response (trastuzumab response proxy) because the path centers on drug efficacy conditional on ERBB2 status and microenvironment rather than long-term outcomes; a therapy-response endpoint directly tests the mechanistic interaction between ERBB2 level and immune context. 5) Deliberate exclusions: I excluded TLS-like aggregates and stromal-to-tumor ratio to keep the hypothesis sharp and focused on the most directly relevant immune imaging feature (TIL density) and the single anchor gene ERBB2; I also did not include additional genes or Fc-receptor polymorphisms (not present in the input) to avoid inventing features?

Working Hypothesis

In triple-negative breast cancers with measurable but low/heterogeneous ERBB2 expression, trastuzumab induces clinically meaningful tumor responses only when the tumor microenvironment is rich in tumor-infiltrating lymphocytes (high TIL density). Mechanistically, trastuzumab-bound ERBB2-expressing tumor cells engage Fc receptors on nearby effector cells (e.g., NK cells and TILs), driving ADCC and tumor cell death; a high TIL density provides sufficient effector cells to mediate this effect, leading to objective responses. In ERBB2-low/heterogeneous tumors with low TIL density, Fc-mediated engagement is inadequate and responses are minimal. ERBB2 copy-number heterogeneity may elevate the probability of ERBB2-expressing clones within a TIL-rich microenvironment, increasing the likelihood of response relative to uniformly ERBB2-low tumors. Clinically, measuring TIL density together with ERBB2 status will stratify TNBC patients who could benefit from trastuzumab despite not meeting classical HER2-overexpression criteria, with higher expected objective response rates in TIL-high/ERBB2-measurable tumors treated with trastuzumab versus those with low TIL density.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	ERBB2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERBB2: ERBB2 expression might still engage HER2-directed mechanisms — why do some ERBB2-expressing TN tumors fail to benefit from trastuzumab while others might respond? 2) Tier 1 path used: the provided path links 'triple-negative phenotype → outcome → trastuzumab → HER2/ERBB2 → breast tissue → breast invasive carcinoma'; I built on ERBB2 as the molecular anchor because trastuzumab efficacy is mechanistically tied to ERBB2 expression.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	HER2_low_expression	Supporting modifier from knowledge graph
Secondary	ERBB2_copy_number_heterogeneity	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective/biomarker analyses correlating TIL density and ERBB2 status with any trastuzumab exposure outcomes in TNBC,

Prediction 2: in vitro ADCC assays using patient-derived ERBB2-expressing tumor cells and autologous effector cells to quantify Fc-dependent killing across TIL densities,

Prediction 3: ex vivo organoid/PDX models incorporating human immune components to measure trastuzumab-driven tumor regression under varying TIL densities and ERBB2 copy-number patterns, and

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- HER2_low_expression: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- ERBB2_copy_number_heterogeneity: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.